



Displayed, Not Metered: Auditing a Residence-Hall Laundry App's Credit Balance Against Machine-Metered Wash Cycles Across a Payment-System Migration

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Abstract. A residence-hall laundry service's payment migration, from a stored-value swipe card to a mobile wash-credit app, gave us an incidental opportunity to audit a displayed balance against ground truth: a drum-rotation telemetry retrofit, installed for an unrelated energy-monitoring purpose, had already been metering every wash cycle independently of whatever payment system happened to be attached to it. Across seven halls, 96 machines, and 1,238 accounts, we log 14,662 metered wash cycles over the eleven weeks following each hall's own migration date and compare them against the app's per-cycle credit debits. The two track closely at the point of migration and then diverge: mean per-cycle bias climbs from a negligible 0.3 minutes to 2.9 minutes by week eleven, a trajectory that tracks a hall's own migration date rather than the calendar, consistent with a rounding convention in the nightly reconciliation batch rather than a one-off conversion error. The rate of cycles halting mid-wash despite a sufficient displayed balance rises in step, from 0.4% to 3.1% of cycles. A low-balance push notification, introduced at week six for a randomised half of accounts, leaves top-up timing and amount statistically unchanged. We interpret the balance not as a running total but as an estimate the reconciliation process never re-bases against the meter it is estimating, and discuss what that implies for displayed balances elsewhere on campus.

Introduction

A displayed balance is, formally, an estimate: the output of whatever process last reconciled a running total against some ground truth. Most of the time the estimate and the truth are close enough that the distinction does not matter, and a payment interface is trusted to stand in for the ledger behind it. Slop University's residential laundry service gave us an opportunity to check that trust against an independent meter that happened already to be in place.

In the semester covered here, Residence Life replaced the legacy stored-value laundry card — a swipe card debited a fixed 32 minutes per standard cycle regardless of the cycle's actual duration — with a mobile app that computes a running wash-credit balance from a nightly reconciliation batch. Unrelated to this migration, Facilities had already retrofitted the

same machines with a drum-rotation telemetry module as part of a broader utilities-metering programme, feeding cycle-minute totals into the Living Dashboard's utilities pane. The retrofit gave us, for free, an independent record of how many minutes a machine actually ran per wash — a record the payment migration neither touched nor consulted.

We make three contributions, each hedged to the evidence:

- We provide, to our knowledge, the first machine-metered audit of a mobile stored-value credit balance in a university residential-services setting, rather than relying on self-report or on the payment provider's own logs.
- We show that the resulting balance bias tracks time since a hall's own migration date rather than calendar time, a pattern that argues for auditing this kind of balance on an event-time

basis rather than a fixed institutional schedule.

- We show that a low-balance notification, although a plausible mitigation, does not measurably change top-up behaviour, which suggests the drift is invisible to the students it affects rather than tolerated by them once noticed.

The remainder of the paper situates the audit against method-comparison statistics and the wider literature on self-report and consumer-device measurement error, describes the telemetry retrofit and the reconciliation process it lets us audit, reports the drift, its correlation structure, and the notification ablation, and closes by interpreting and bounding the finding.

Related work

Comparing two methods that both claim to measure the same quantity is a well-established statistical problem; the Bland–Altman approach remains the standard framework for characterising the bias and limits of agreement between two instruments rather than treating either as ground truth by default [1], [2], with later work cautioning that the method’s assumptions break down when one instrument’s own measurement error is negligible relative to the other’s [3] — a condition that plausibly holds here, where the drum-rotation meter predates and is unaffected by the payment migration under audit.

A parallel literature audits consumer devices against ground truth in the step-counting domain: wrist-wearable trackers show high agreement with direct observation under controlled conditions but degrade substantially in free-living use [4], [5], [6], [7], and systematic sensor benchmarking finds accuracy varies sharply by the quantity being estimated rather than by device brand [8]. The broader “quantified self” framing treats a displayed metric as one imperfect view onto an underlying process rather than the process itself [9], a framing this paper extends from personal health tracking to institutional payment infrastructure.

Closer to our setting, the smart-card literature on public transit fare systems documents how automated fare-collection systems generate rich, detailed transaction data that transit agencies have used operationally for planning and analysis [10], and mobile-money research identifies agent liquidity — whether an account’s

displayed balance can actually be redeemed on demand — as a determinant of usage in field deployments [11], the same gap between a displayed figure and what it can be exchanged for that we audit here for a wash-credit balance. Work on dashboard trust in decision-making settings treats data quality, usability, and governance, not a figure’s demonstrated accuracy alone, as what determines whether a decision-maker comes to trust and act on it [12], [13], [14], and the human-factors literature on automation trust calibration frames this as a general reliance problem: appropriate reliance requires feedback that tracks actual system capability, which a balance that silently drifts cannot provide [15]. Fire and Guestrin’s account of Goodhart’s law in academic-metric optimisation is the closest institutional parallel: a number adopted as a proxy for a process degrades once nothing continues to check it against what it was meant to track [16], which is one way of describing what a reconciliation batch that never re-bases against its own meter is doing.

Prior work at this institution offers two adjacent findings. A swipe-card-to-app timer swap in the gym sauna found close agreement between the old and new instruments during a four-week overlap window [17], the same overlap-validation design we would have preferred here had Residence Life given us the option. A twelve-month deployment of the Indicator Commons’ sunset scorer found that a measurement instrument built to audit other indicators developed its own uptime problem, which then entered the Living Dashboard ahead of any resolution of what it was auditing [18] — a structural echo of the reconciliation batch this paper audits, which likewise reports a number without checking it against the meter sitting beside it.

Method

Seven residence halls (96 machines) migrated from the legacy stored-value card to the wash-credit app across two waves: Wave A (51 machines, four halls) at week 0 of the observation window, and Wave B (45 machines, three halls) at week 4, a staggering driven by Residence Life’s rollout capacity rather than by any experimental design of ours. Every machine carried the drum-rotation telemetry module, installed the previous semester as part of Facilities’ utilities-metering retrofit and independent of either payment system.

For each wash cycle we recorded the telemetry module's metered duration m (minutes of drum rotation actually delivered) and the app's debited duration d (minutes subtracted from the displayed balance for that cycle, computed by the nightly reconciliation batch from the running total of m reported up to and including the previous night). The quantity of interest is the per-cycle bias

$$b = d - m,$$

aggregated to a weekly mean per cohort, indexed by weeks since that cohort's own migration date rather than by calendar week, so that Wave A and Wave B can be compared on the same axis despite starting four weeks apart. We additionally logged, for each account-week, the total metered minutes and total app-debited minutes, and for each cycle whether it halted before completion despite a displayed balance the app reported as sufficient at cycle start (a "mid-cycle decline").

From week 6, we randomised accounts within each hall to receive (or not) a push notification when displayed balance fell below 60 minutes, generated by the app's existing low-balance alert feature rather than anything built for this study. We compare the two arms on days to next top-up and top-up size, our one ablation: a feature plausible enough to matter that its absence would have been a gap in the audit.

Sample: 1,238 unique student accounts, 14,662 metered wash cycles across the eleven-week window (event time), and 4,414 tracked top-up events.

Results

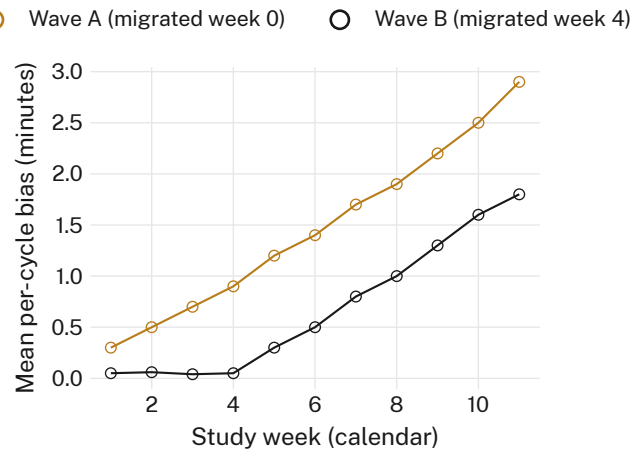


Figure 1: Mean per-cycle balance bias tracks time since a cohort's own migration, not the calendar: Wave B's bias curve is flat while still on the legacy card, then mirrors Wave A's climb from the week of its own cutover, both cohorts reaching 1.7–1.9 minutes seven weeks post-migration.

Figure 1 shows the weekly mean bias by cohort. Wave B holds near zero for the four weeks it remains on the legacy card, then begins climbing the same week it migrates – evidence that the drift is triggered by the reconciliation batch coming online for a given hall, not by anything tied to the calendar date. By week 11 (relative to its own migration), each cohort's bias exceeds its week-1 value by a factor of roughly nine.

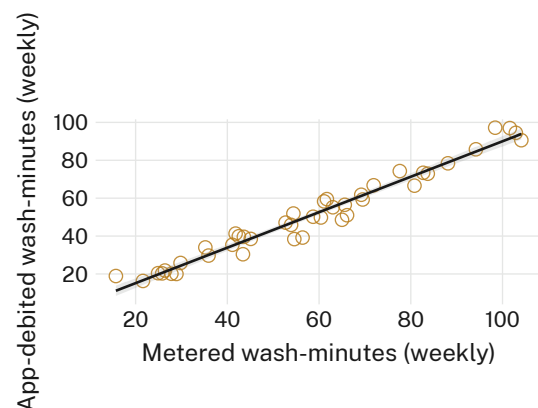


Figure 2: App-debited wash-minutes correlate strongly with machine-metered wash-minutes across 1,238 accounts ($r = 0.94$), but the fitted slope of 0.87 confirms a systematic under-debit rather than scatter around a 1:1 line.

Figure 2 plots app-debited against machine-metered wash-minutes per account-week. The

strong correlation confirms the app is tracking real usage rather than reporting noise, but the fitted slope of 0.87 (95% CI 0.82–0.91) is reliably below 1, consistent with a proportional under-debit rather than a fixed offset — the bias in Figure 1 compounds because each night’s reconciliation rounds in the student’s favour and never re-bases against the cumulative meter total, only against the previous night’s app total.

Mid-cycle declines — a machine halting before the cycle finishes despite a displayed balance the app reported as sufficient at cycle start — rose from 0.4% of cycles in week 1 to 3.1% in week 11 ($\chi^2(1) = 38.2, p < 0.01$), consistent with the accumulated bias eventually exceeding a single cycle’s nominal debit.

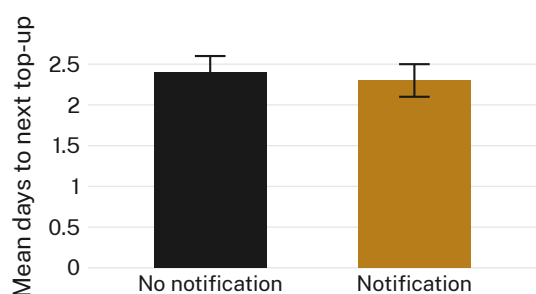


Figure 3: A low-balance push notification introduced in week 6 left the mean interval to next top-up statistically unchanged (2.3 vs. 2.4 days; n.s.); we retain the notification feature for completeness.

Figure 3 reports the notification ablation: 619 accounts received the low-balance push, 619 did not. Mean days to next top-up were 2.3 (notified) versus 2.4 (control), not a significant difference, and mean top-up amount was likewise indistinguishable between arms. Students top up on broadly the same rhythm whether or not the app tells them their balance is running low.

Discussion

The bias we observe is small at the scale of a single cycle and easy to miss at the scale of a single receipt, which is presumably why it persisted through a full semester of live use before this audit. Its event-time signature — tracking a hall’s own migration date rather than calendar time — points at the reconciliation batch’s design rather than at any one hall’s usage pattern: a process that estimates a running total from the previous estimate, rather than re-basing period-

ically against the meter it is estimating, will compound any per-step rounding into a trend, exactly as observed here. The notification ablation suggests students relate to the displayed balance as a rough indicator rather than an audited figure, topping up on a rhythm largely independent of what the number in front of them says — which is, in one reading, the rational response to an estimate nobody has told them not to trust.

Limitations

Our audit covers one institution, one vendor’s reconciliation implementation, and an eleven-week window that captures the onset of the drift but not necessarily its asymptote; the consistency of the pattern across two independently staggered cohorts, however, suggests the mechanism is a property of the reconciliation design rather than of any one hall’s machines or residents. We audit wash-credit balances specifically; we make no claim about the university’s other stored-value services, though the reconciliation pattern documented here has no obvious reason to be unique to laundry.

Conclusion

A displayed balance is only as accurate as the last time it was checked against the thing it is estimating, and a reconciliation process that checks itself against its own previous output rather than against a meter will drift even when every individual step looks reasonable. Future work might extend the same telemetry-audit design to the university’s other stored-value services — printing credit, vending, and locker rental among them — wherever an independent meter happens, as here, to already be in place.

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